



LORENTZ CENTER WORKSHOP

Accelerating Astrophysical Discovery

with Foundation Models

Participant booklet

14–18 September 2026 · Leiden, The Netherlands

VERSION 1.1 · 14 SEPTEMBER 2026

WELCOME

A week built for making

Dear Lorentz Workshop participant,

This booklet introduces the consortium's ambition and relevancy graph, lays out the working timetable, and suggests a few things to bring into the week. Bring concrete examples, informed questions, and a willingness to build across disciplinary boundaries.

AT A GLANCE

Workshop

Accelerating Astrophysical Discovery with Foundation Models

Dates

Monday 14 to Friday 18 September 2026

Venue

Lorentz Center@omega, Leiden, The Netherlands

Start times

Monday at 10:00; Tuesday–Friday at 9:30

CONTINUE WITH THE CONSORTIUM

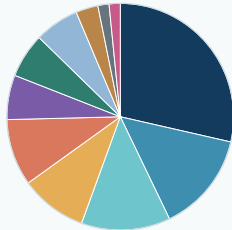
Visit the [consortium website](#) for workshop materials, news, and ongoing work. Interested in taking part? [Join the consortium](#) through the same sign-up form linked from the members page.

WORKSHOP COMMUNITY

List of participants

Last name	First name	Affiliation
Albert	Joshua G.	California Institute of Technology
Albin	Thomas	Freelancer Scientist
Angeloudi	Eirini	University of Hamburg, Germany
Bester	Hertzog Landman	SARAO, South Africa
Bhihe	Cedric	BSC, Spain
Callingham	Joseph	ASTRON/UvA
Cañas-Herrera	Guadalupe	Leiden Observatory
Caron	Sascha	Radboud University
Cheng	Ting-Yun	University of Groningen
Cluca	Ioana	Stanford University, USA
Cruz	Marcos	Instituto de Física de Cantabria, Spain
Cuesta Lazaro	Carolina	Flatiron Institute / New York University
De la Torre Luque	Pedro	Universidad Autónoma de Madrid (UAM)
Eckner	Christopher	Instituto de Astrofísica de Canarias
Fernández Sopena	Carlos	ICE/CSIC, Spain
Hartman	Nicole	MIT / IAIFI
Heng	Ik Siong	University of Glasgow, UK
Huertas-Company	Marc	IAC
Huppenkothen	Daniela	University of Amsterdam, The Netherlands
Jain	Divesh	University of Heidelberg
Keenan	Ryan	Max Planck Institute for Astronomy
Keller	Christoph	National Solar Observatory
Khan	Sebastian	Starling Bank
Leonhard	David	University of Bonn
Li	Yyouyou	GRAPPA, University of Amsterdam
Lopez	Melissa	Nikhef
Loukili	Soulaymane	BSC, Spain
Madreddy	Sandeep	Simons SkAI, Argonne National Laboratory
Pérez Romero	Judit	IFIC (UV/CSIC)
Portgies Zwart	Simon	Leiden Observatory
Poux-Bouret	Benjamin	LIRA, Observatoire de Paris, PSL, Sorbonne, Paris Cité, CY Cergy Paris Université, CNRS
Ruiz de Austri	Roberto	Universidad de Valencia
Sáinz-Pardo Díaz	Judith	Spanish National Research Council (CSIC) – Institute of Physics of Cantabria (IFCA)
Samia	Noelle	Simons SkAI
Smirnov	Oleg	Rhodes University & SARAO
Smith	Michael	Harvard-Smithsonian Center for Astrophysics
Sotoudeh	Hadi	University of Cambridge
Stevance	Héloïse	Oxford University, UK
Tallada Crespi	Pau	CIEMAT-PIC
Vacheret	Antonin	Laboratoire Corpusculaire de Caen UMR 6534, Caen, France
Villaescusa-Navarro	Francisco	Simons Foundation
Weigel	Philip	Harvard University
Zaharijas	Gabrijela	University of Nova Gorica
Zhao	Yue	SURF, NL

Consortium fields



Share of field entries

AI	28.6%
Cosmology	14.3%
Radio	12.7%
Computational physics	9.5%
Gravitational waves	9.5%
Gamma-rays	6.3%
Neutrinos	6.3%
Optical	6.3%
X-Ray	3.2%
CRs	1.6%
Infrared	1.6%

Two connected pillars

PILLAR 01

Design science for human–AI collaboration

The first goal is to design how future astrophysicists and cosmologists will do science in collaboration with AI. This is more than adding chat interfaces to existing tools. It means redesigning workflows so humans, reasoning language models, simulators, archives, instruments, and foundation models can jointly form hypotheses, retrieve evidence, run checks, update beliefs, and communicate results.

PILLAR 02

Build a joint representation of physical data

The second goal is to build the technical substrate: a model that learns across observations, simulations, synthetic observations, language, code, and instrumental context. It should support meaningful embeddings for retrieval and serendipity, along with generative capabilities for conditional inference across linked views of the same physical system.

One shared ambition. The two pillars meet in an agentic discovery system: reasoning models guide scientific work, the joint-representation model grounds that work in physical evidence, and humans remain responsible for judgement.

TRAINING STRUCTURE

The relevancy graph



FROM DATA RELATIONSHIPS TO ATTENTION

We envision a large graph database whose nodes are physical data and whose links encode relevancy. An astrophysical system, for example, can be observed repeatedly by different instruments in different modalities and simulated by different codes.

During self-supervised training, sampled graph connectivity forms the attention mask. Text naturally supplies an autoregressive order; physical data admit a more general attentional field—cross-modal, many-to-many, and hierarchical. The model uses this context to predict or reconstruct withheld data.

Nodes are physical data. Relevancy links define which data may exchange information during training.

BEFORE LEIDEN

Bring ingredients, not finished answers

ASTROPHYSICS + COMPUTATIONAL PHYSICS

Think about use cases where AI could help humans study the cosmos and do better science if it could understand, reason about, and generate physical data. Prepare a list of astrophysical data you can access. Any modality is welcome, including:

- observations, catalogues, spectra, images, time series, and event lists;
- simulation code, snapshots, initial conditions, and synthetic observations;
- multimodal observations of the same systems;
- instrument responses, calibration context, uncertainties, and provenance.

AI + COMPUTATIONAL INFRASTRUCTURE

Think about the architectural challenges facing a joint-representation model that must learn deep, self-supervised representations of large physical datasets. Also consider the infrastructure required to train across physical datasets that can dwarf the data used to train language models:

- long context, heterogeneous tokenisation, and shared latent spaces;
- compute-to-data training across distributed archives and HPC systems;
- data access, preprocessing, provenance, monitoring, and validation;
- staged scaling, simulator-backed synthetic data, and failure modes.

If you span both groups, bring both perspectives. If you span neither exactly, bring the constraints, examples, and questions from your own domain.

THE WEEK

The week at a glance

Day	Focus	Main deliverable
Monday	Vocabulary, use cases, evaluation	Use cases and evaluation criteria
Tuesday	Parallel work: data + model	First tests of data and model hypotheses and interfaces
Wednesday	Parallel work: data + model	Reviewed evidence and hypotheses ready to implement
Thursday	AI-assisted hypothesis implementation	Working hypotheses implemented in code with AI assistance
Friday	Roadmap, governance, funding	Roadmap

THE HAND-OFF

Monday establishes the scientific use cases for everyone. On Tuesday and Wednesday, technical work proceeds in two parallel tracks: **Data** (relevancy graph, representation, and tokenisation) and **Model** (architecture, training, and physical alignment). Wednesday reviews evidence and prepares the hypotheses for implementation. Thursday implements working hypotheses in code with AI assistance. Friday turns the week's results into a roadmap.

Formal working hours are 10:00–16:30 on Monday, 9:30–17:00 Tuesday–Thursday, and 9:30–13:30 on Friday. The building is open for arrivals and coffee from 09:00 on Monday.

WORKING TOGETHER

Consortium workstreams

Workstream	Working focus
Scientific use cases	Workflows, required functionality, and evaluation targets.
Data, relevancy graph, tokenisation, and representation	Linked data, typed relationships, provenance, and encodings that preserve physical information.
Architecture, training, and physical alignment (RLSF)	Model hypotheses, learning objectives, generated states, simulation feedback, and discriminating tests.
Infrastructure and prototyping	Data-local compute, sampling, software, and runnable components.
Project oversight	Dependencies, decisions, priorities, milestones, and ownership.
Network and funding	Expertise, partners, resources, funding, and sustainable participation.

Stay connected Share artifacts, evidence, open questions, and requests in plenary. Use public GitHub Issues for non-sensitive coordination and decisions. Human-review AI summaries; keep credentials and restricted information private.

MONDAY · 10:00–16:30

Timetable

Purpose of the day

Define scientific use cases and evaluation criteria that guide the rest of the week.

Time	Session
09:00–10:00	Arrival and coffee
10:00–10:30	Kickoff talk
10:30–12:00	Science 2.0 and use cases
12:00–13:30	Lunch
13:30–14:30	Group synthesis
14:30–15:00	Coffee break
15:00–16:30	Evaluation targets and workstream planning
16:30 onward	Welcome reception

Keep asking What would count as evidence that this workflow produces better science, rather than simply more output?

TUESDAY · 9:30–17:00

Timetable

Purpose of the day

Develop and test hypotheses in two parallel tracks: **Data** (relevancy graph, representation, and tokenisation) and **Model** (architecture, training, and physical alignment).

Time	Session
9:30–10:00	Vocabulary and hypotheses kickoff
10:00–10:30	Parallel workstreams
10:30–11:00	Coffee break
11:00–12:00	Parallel workstreams
12:00–13:30	Lunch
13:30–14:30	Group synthesis
14:30–15:00	Coffee break
15:00–17:00	Parallel workstreams

Keep asking What could we make or test now, and what does another team need from us?

WEDNESDAY · 9:30–17:00

Timetable

Purpose of the day

Review the evidence, continue the parallel Data and Model work, and prepare selected hypotheses for implementation.

Time	Session
9:30–10:00	Progress and experimental-results review
10:00–10:30	Parallel workstreams
10:30–11:00	Coffee break
11:00–12:00	Parallel workstreams
12:00–13:30	Lunch
13:30–14:30	Group synthesis
14:30–15:00	Coffee break
15:00–17:00	Parallel workstreams
17:00–20:00	Workshop dinner

THURSDAY · 9:30–17:00

Timetable

Purpose of the day

Implement selected hypotheses in code with AI assistance and test what the implementations demonstrate.

Time	Session
9:30–10:00	Organiser briefing and overnight review
10:00–10:30	Implementation planning
10:30–11:00	Coffee break
11:00–12:00	AI-assisted implementation I
12:00–13:30	Lunch
13:30–14:30	AI-assisted implementation II
14:30–15:00	Coffee break
15:00–16:15	Show-and-tell and integration
16:15–17:00	Closeout and Friday handoff

Thursday rule: every artifact should connect to a use case, a model capability, and an evaluation target.

FRIDAY · 9:30–13:30

Timetable

Purpose of the day

Turn the week’s results, open questions, ownership, and resource needs into a roadmap.

Time	Session
9:30–10:00	Review of the week
10:00–10:30	Roadmap group discussion
10:30–11:00	Coffee break
11:00–11:30	Funding opportunities
11:30–12:00	Roadmap synthesis
12:00–13:30	Closing lunch

PRACTICAL MATTERS

Discounted hotel reservations are limited, so please book as soon as possible if you need accommodation.

For practical questions or help with accommodation, contact Maria Krebbers at acceleratingastrophysical@lorentzcenter.nl.

accelerating-astrophysical-discovery.github.io

Bring a use case. Bring a constraint. Leave with something we can build together.